

The WKB Approximation and Adiabatic Regularization

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Introduction

Conventions We use the mostly plus metric signature, i.e. $\eta_{\mu\nu} = (-, +, +, +)$ and units where $c = \hbar = k_B = 1$. The reduced four dimensional Planck mass is $M_{\text{Pl}} = (8\pi G)^{-1/2} \approx 2.43 \times 10^{18} \text{ GeV}$. The d'Alembert and Laplace operators are defined to be $\square \equiv \partial_\mu \partial^\mu = -\partial_t^2 + \nabla^2$ and $\nabla^2 = \partial_i \partial^i$ respectively. We use boldface letters $\mathbf{r}$ to indicate 3-vectors and x and p to denote 4-vectors. Conventions for the curvature tensors, covariant and Lie derivatives are all taken from Carroll.

The WKB Approximation

First we expand out the $\square$ operator

$$\partial_t^2 \varphi + (d-1) \frac{\dot{a}}{a} \partial_t \varphi - \frac{\partial_i \partial^i \varphi}{a^2} + M^2(t) \varphi + \xi R \varphi = 0 \quad (1)$$

we can make the usual ansatz

$$\varphi(\mathbf{r}, t) = \frac{1}{\sqrt{\mathcal{V}}} \sum_{\mathbf{p}} [a_{\mathbf{p}} u_{\mathbf{p}}(\mathbf{r}, t) + a_{\mathbf{p}}^\dagger u_{\mathbf{p}}^*(\mathbf{r}, t)], \quad (2)$$

where $\mathcal{V}$ is the volume of the box that we quantize our field in. We will further make the following simplification¹ in order to eliminate the term that has one time derivative

¹Note: if we were working with a more general global curvature k , then we would introduce into our ansatz a term that looks like $Y_{\mathbf{p}}(\mathbf{r})$ which satisfies $\nabla^2 Y_{\mathbf{p}} = -(\mathbf{p}^2 - k)Y_{\mathbf{p}}$. It can be readily seen that in a flat geometry, $Y_{\mathbf{p}}(\mathbf{r}) = e^{i\mathbf{p} \cdot \mathbf{r}}$.

$$u_{\mathbf{p}}(\mathbf{r}, t) = \frac{f_{\mathbf{p}}(t)e^{i\mathbf{p}\cdot\mathbf{r}}}{a^{\frac{d-1}{2}}(t)}, \quad (3)$$

which we can compute its various time derivatives

$$\partial_t u_{\mathbf{p}}(\mathbf{r}, t) = \frac{\dot{f}_{\mathbf{p}}e^{i\mathbf{p}\cdot\mathbf{r}}}{a^{\frac{d-1}{2}}} - \frac{d-1}{2} \frac{\dot{a}}{a} u_{\mathbf{p}}, \quad (4)$$

$$\partial_t^2 u_{\mathbf{p}} = \frac{\ddot{f}_{\mathbf{p}}e^{i\mathbf{p}\cdot\mathbf{r}}}{a^{\frac{d-1}{2}}} - (d-1) \frac{\dot{a}}{a} \frac{\dot{f}_{\mathbf{p}}e^{i\mathbf{p}\cdot\mathbf{r}}}{a^{\frac{d-1}{2}}} - \frac{d-1}{2} \frac{\ddot{a}}{a} u_{\mathbf{p}} + \frac{d^2-1}{4} \frac{\dot{a}^2}{a^2} u_{\mathbf{p}}. \quad (5)$$

This brings the EoM to the form

$$\ddot{u}_{\mathbf{p}} + (d-1) \frac{\dot{a}}{a} \dot{u}_{\mathbf{p}} - \frac{\nabla^2 u_{\mathbf{p}}}{a^2} = \frac{\ddot{f}_{\mathbf{p}}e^{i\mathbf{p}\cdot\mathbf{r}}}{a^{\frac{d-1}{2}}} - \frac{(d-1)(d-3)}{4} \frac{\dot{a}^2}{a^2} \frac{f_{\mathbf{p}}e^{i\mathbf{p}\cdot\mathbf{r}}}{a^{\frac{d-1}{2}}} - \frac{d-1}{2} \frac{\ddot{a}}{a} \frac{f_{\mathbf{p}}e^{i\mathbf{p}\cdot\mathbf{r}}}{a^{\frac{d-1}{2}}} + \frac{\mathbf{p}^2}{a^2} \frac{f_{\mathbf{p}}e^{i\mathbf{p}\cdot\mathbf{r}}}{a^{\frac{d-1}{2}}}. \quad (6)$$

Putting all the pieces together leaves us with

$$\ddot{f}_{\mathbf{p}} - \frac{(d-1)}{2} \frac{\ddot{a}}{a} f_{\mathbf{p}} - \frac{(d-1)(d-3)}{4} \frac{\dot{a}^2}{a^2} f_{\mathbf{p}} + \frac{\mathbf{p}^2}{a^2} f_{\mathbf{p}} + M^2(t) f_{\mathbf{p}} + \xi R f_{\mathbf{p}} = 0, \quad (7)$$

or if we plug in the Ricci scalar's form in terms of the d -dimensional FRW metric

$$\ddot{f}_{\mathbf{p}} + \omega_{\mathbf{p}}^2 f_{\mathbf{p}} + \left[2\xi(d-1) - \frac{d-1}{2} \right] \frac{\ddot{a}}{a} f_{\mathbf{p}} + \left[\xi(d-1)(d-2) - \frac{(d-1)(d-3)}{4} \right] \frac{\dot{a}^2}{a^2} f_{\mathbf{p}} = 0, \quad (8)$$

where we've defined another variable

$$\sigma(t) = 3\lambda\phi_0^2 + (d-1) \left[2\xi - \frac{1}{2} \right] \frac{\ddot{a}}{a} + (d-1) \left[\xi(d-2) - \frac{(d-3)}{4} \right] \frac{\dot{a}^2}{a^2}, \quad (9)$$

which reduces our EoM to being

$$\ddot{f}_{\mathbf{p}}(t) + [\omega_{\mathbf{p}}^2(t) + \sigma(t)] f_{\mathbf{p}}(t) = 0, \quad (10)$$

which is the familiar form of a harmonic oscillator with a time dependent frequency. Because the only dependence on $\mathbf{p}$ enters in via its magnitude, we can move forward with the knowledge that the mode functions will only depend on the magnitude and thus we can drop the vector subscripts we've been using thus far. To deal with these equations

of motion for the fluctuation, it's typical to introduce the WKB approximation where we assume the universe's expansion is sufficiently slow

$$f_{\mathbf{p}}(t) = \frac{\exp\left(-i \int^t W_p(t') dt'\right)}{\sqrt{2W_p(t)}}. \quad (11)$$

Acting time derivatives on the mode functions in this approximation yields

We make the following ansatz:

$$\dot{f}_p(t) = -\left[iW_p(t) + \frac{\dot{W}_p(t)}{2W_p(t)}\right]f_p(t). \quad (12)$$

We also have

$$\ddot{f}_p(t) = \left[\frac{\dot{W}_p(t)}{2W_p(t)} + iW_p(t)\right]^2 f_p(t) - \left[i\dot{W}_p(t) + \frac{\ddot{W}_p(t)}{2W_p(t)} - \frac{\dot{W}_p^2(t)}{2W_p^2(t)}\right]f_p(t). \quad (13)$$

Plugging this into the EoM for the mode functions gives

$$W_p^2(t) = \omega_p^2(t) + \sigma(t) - \frac{1}{2}\left[\frac{\ddot{W}_p(t)}{W_p(t)} - \frac{3}{2}\left(\frac{\dot{W}_p(t)}{W_p(t)}\right)^2\right]. \quad (14)$$

Because we are left with the above expression to solve, the only way to extract analytic results are to do perturbation theory recursively i.e. first solve for the 0-th adiabatic order term, plug that in to find the 2nd adiabatic order (and it's second because we have to square the 0-th) and so on. Here, the “expansion parameter” are the number of time derivatives. N -th derivatives of ω_p are referred to as being of n -th adiabatic order. One way to see this as well as getting a better understanding of the ‘slowly varying’ condition is by replace $t \rightarrow t/T$, where T denotes the adiabatic order. This brings our Schrodinger equation to the form

$$\frac{d^2 f_p}{d\bar{t}^2} + T^2 \omega_p^2(\bar{t}) f_p(\bar{t}) = 0, \quad (15)$$

where $\bar{t} = t/T$ and hence

$$\frac{d}{dt} a^2(t/T) = \frac{1}{T} \frac{d}{d\bar{t}} a^2(\bar{t}). \quad (16)$$

Thus, we can see that when we take the $T \rightarrow \infty$ limit, $a^2(\bar{t})$ will vary quite slowly. The literature often refers to having a, T^{-n} , n-th order adiabatic approximation to denote how many derivatives of ω_p and by extension a^2 one has taken. consequently, terms like ω_p are considered 0-th order in the adiabatic expansion. We present the adiabatic order for each term that appears above

$$[\omega_p]_{\text{ad}} = 0, \quad [H]_{\text{ad}} = 1, \quad [R]_{\text{ad}} = [\phi_0^2]_{\text{ad}} = 2, \quad (17)$$

and all time derivatives increases the adiabatic order of a term by one. We regard the term ϕ_0^2 as 2nd adiabatic order because only two derivatives of ϕ_0 are required to cancel out with the divergences that creep up in the stress energy tensor which is a 4th-order adiabatic term.

Overall, this is the prescription for solving for the mode functions in a generally (slowly-) varying spacetime. This framework also lends itself well with our notion of what constitutes a particle. However, there are different incidences where this approximation breaks down. First we note that one can solve the WKB equation iteratively. To zeroth-order we have

$$W_p^{(0)} \simeq \omega_p. \quad (18)$$

At second order, we have

$$\begin{aligned} (W_p^{(2)})^2 &\simeq \omega_p^2 + \sigma(t) - \frac{1}{2} \left[\frac{\ddot{W}_p^{(0)}(t)}{W_p^{(0)}(t)} - \frac{3}{2} \left(\frac{\dot{W}_p^{(0)}(t)}{W_p^{(0)}(t)} \right)^2 \right] \\ &= \omega_p^2(t) + \sigma(t) - \frac{1}{2} \left[\frac{\ddot{\omega}_p(t)}{\omega_p(t)} - \frac{3}{2} \left(\frac{\dot{\omega}_p(t)}{\omega_p(t)} \right)^2 \right]. \end{aligned} \quad (19)$$

The condition for the adiabatic approximation to hold is then

$$\left(\frac{\dot{\omega}_p(t)}{\omega_p(t)} \right)^2 \ll 1. \quad (20)$$

For now, we will take $\phi_0 = 0$. Let's derive the 4-th order adiabatic wave function. First we notice that the first and second time derivatives of the frequency can be written as

$$\dot{\omega}_p = -\frac{\dot{a}p^2}{a^3\omega_p} = -\frac{(\omega_p^2 - m^2)H}{\omega_p}, \quad (21)$$

$$\ddot{\omega}_p = \frac{2p^2}{a^2} \frac{H^2}{\omega_p} - \frac{p^2}{a^2} \frac{\dot{H}}{\omega_p} - \frac{p^4}{a^4} \frac{H^2}{\omega_p^3} = \frac{3m^2 H^2}{\omega_p} \left(1 - \frac{m^2}{\omega_p^2}\right) - \frac{\omega_p^2 - m^2}{\omega_p} (\dot{H} + H^2). \quad (22)$$

Rewriting the σ term as

$$\sigma(t) = \left(\xi - \frac{1}{6}\right)R - \frac{1}{2}\left[\dot{H} + \frac{1}{2}H^2\right], \quad (23)$$

with these terms in hand, we can compute the second order WKB wave function

$$\begin{aligned} W_p^{(2)} &= \sqrt{\omega_p^2 + \left(\xi - \frac{1}{6}\right)R - \frac{1}{2}\left[\dot{H} + \frac{1}{2}H^2\right] + \frac{3}{4}\frac{\dot{\omega}_p^2}{\omega_p^2} - \frac{\ddot{\omega}_p}{2\omega_p}} \\ &= \omega_p \sqrt{1 + \frac{(\xi - \frac{1}{6})R}{\omega_p^2} - \frac{2\dot{H} + H^2}{4\omega_p^2} + \frac{3\dot{\omega}_p^2}{4\omega_p^4} - \frac{\ddot{\omega}_p}{2\omega_p^3}} \\ &\simeq \omega_p + \frac{(\xi - \frac{1}{6})R}{2\omega_p} - \frac{2\dot{H} + H^2}{8\omega_p} + \frac{3\dot{\omega}_p^2}{8\omega_p^3} - \frac{\ddot{\omega}_p}{4\omega_p^2}. \end{aligned} \quad (24)$$

The last two terms that appear in the second order adiabatic contribution can be seen as

$$\frac{3}{4}\frac{\dot{\omega}_p^2}{\omega_p^2} - \frac{\ddot{\omega}_p}{2\omega_p^2} = \frac{H^2}{8\omega_p} - \frac{3m^2 H^2}{4\omega_p^3} + \frac{5m^4 H^2}{8\omega_p^5} + \frac{\dot{H}}{4\omega_p} - \frac{m^2 \dot{H}}{4\omega_p}. \quad (25)$$

This yields the following term form for $W_p^{(2)}$,

$$W_p^{(2)} \simeq \omega_p + \frac{(\xi - \frac{1}{6})R}{2\omega_p} - \frac{m^2}{4\omega_p^3}(\dot{H} + 3H^2) + \frac{5m^4}{8\omega_p^5}H^2, \quad (26)$$

and we note that each term (with the exception of the very first term) is at most 2nd order in the adiabatic WKB series. The time derivative of the above term becomes

$$\begin{aligned} \dot{W}_p^{(2)} &\simeq \dot{\omega}_p - \frac{(\xi - \frac{1}{6})\dot{\omega}_p R}{2\omega_p^2} + \frac{(\xi - \frac{1}{6})\dot{R}}{2\omega_p} - \frac{m^2}{4\omega_p^3}(\ddot{H} + 6\dot{H}H) \\ &\quad + \frac{3m^2\dot{\omega}_p}{4\omega_p^4}(\dot{H} + 3H^2) + \frac{5m^4}{4\omega_p^5}\dot{H}H - \frac{25m^4\dot{\omega}_p}{8\omega_p^6}H^2 \\ &\simeq \frac{(m^2 - \omega_p^2)H}{\omega_p} + \frac{(\xi - \frac{1}{6})\dot{R}}{2\omega_p} + \frac{(\xi - \frac{1}{6})(\omega_p^2 - m^2)RH}{2\omega_p^3} - \frac{m^2}{4\omega_p^3}(\ddot{H} + 6\dot{H}H) \\ &\quad - \frac{3m^2(\omega_p^2 - m^2)H(\dot{H} + 3H^2)}{4\omega_p^5} + \frac{5m^4\dot{H}H}{4\omega_p^5} + \frac{25m^4(\omega_p^2 - m^2)H^3}{8\omega_p^7}. \end{aligned} \quad (27)$$

We also need to compute the inverse of $W_p^{(2)}$

$$\begin{aligned}\frac{1}{W_p^{(2)}} &= \frac{1}{\omega_p} \frac{1}{1 + \frac{(\xi - \frac{1}{6})R}{2\omega_p^2} - \frac{2\dot{H} + H^2}{4\omega_p^2} + \frac{3\dot{\omega}_p^2}{4\omega_p^4} - \frac{\ddot{\omega}_p}{2\omega_p^3}} \\ &\simeq \frac{1}{\omega_p} - \frac{(\xi - \frac{1}{6})R}{2\omega_p^3} + \frac{m^2}{4\omega_p^5}(\dot{H} + 3H^2) - \frac{5m^4 H^2}{8\omega_p^7}.\end{aligned}\quad (28)$$

Therefore the ratio $\dot{W}_p^{(2)}/W_p^{(2)}$, which carries up to adiabatic order 3

$$\begin{aligned}-\frac{\dot{W}_p^{(2)}}{W_p^{(2)}} &= \left(1 - \frac{m^2}{\omega_p^2}\right)H - \frac{(\xi - \frac{1}{6})}{2\omega_p^2}(\dot{R} + 2RH) + \frac{(\xi - \frac{1}{6})m^2}{\omega_p^4}RH \\ &\quad + \frac{m^2}{4\omega_p^4}(\ddot{H} + 10\dot{H}H + 12H^3) - \frac{9m^4}{4\omega_p^6}H(\dot{H} + 3H^2) + \frac{15m^6}{4\omega_p^8}H^3.\end{aligned}\quad (29)$$

We also need the second derivative of the 2nd order adiabatic term as well as the ratio between this term and the 2nd order adiabatic term

$$\begin{aligned}\ddot{W}_p^{(2)} &= \frac{(\omega_p^2 - m^2)}{\omega_p^3}H^2 - \frac{(\omega_p^2 - m^2)}{\omega_p}\dot{H} + \frac{(\xi - \frac{1}{6})}{2\omega_p}(\ddot{R} + 2\dot{R}H + R\dot{H} + RH^2) \\ &\quad - \frac{(\xi - \frac{1}{6})m^2}{2\omega_p^3}(R\dot{H} + 2\dot{R}H + 2RH^2) + \frac{3(\xi - \frac{1}{6})m^4}{2\omega_p^5}RH^2 \\ &\quad - \frac{m^2}{4\omega_p^3}(\dot{\dot{H}} + 72H\ddot{H} + 54H^2\dot{H} + 9\dot{H}^2 + 27H^4) + \frac{m^4}{8\omega_p^5}(66H\ddot{H} + 163H^2\dot{H} + 16\dot{H}^2 + 269H^4) \\ &\quad - \frac{5m^6}{8\omega_p^7}H^2(31\dot{H} + 78H^2) + \frac{175m^8}{8\omega_p^9}H^4.\end{aligned}\quad (30)$$

$$\begin{aligned}\frac{\ddot{W}_p^{(2)}}{W_p^{(2)}} &= \left(1 - \frac{m^2}{\omega_p^2}\right)\left[\left(1 + \frac{m^2}{\omega_p^2}\right)H^2 - \dot{H}\right] + \frac{(\xi - \frac{1}{6})}{2\omega_p^2}(\ddot{R} + 2(\dot{R}H + R\dot{H})) \\ &\quad - \frac{(\xi - \frac{1}{6})}{\omega_p^4}(\dot{R}H + R\dot{H}) - \frac{2(\xi - \frac{1}{6})}{\omega_p^2}\left(1 - \frac{m^2}{\omega_p^2}\right)RH^2 \\ &\quad - \frac{m^2}{4\omega_p^4}\left(\dot{\dot{H}} + 12H\ddot{H} + 56H^2\dot{H} + 10\dot{H}^2 + 24H^4\right) - \frac{9m^6}{4\omega_p^8}H^2(9\dot{H} + 22H^2) \\ &\quad + \frac{m^4}{4\omega_p^6}\left(11H\ddot{H} + 9\dot{H}^2 + 4H^2(137\dot{H} + 33H^2)\right) + \frac{45m^8}{2\omega_p^{10}}H^4.\end{aligned}\quad (31)$$

Thus, the 4th-order adiabatic correction is simply

$$\begin{aligned}
W_p^{(4)} = & W_p^{(2)} - \frac{(\xi - \frac{1}{6})}{8\omega_p^3}(\ddot{R} + 5\dot{R}H + 2R\dot{H} + 6RH^2) - \frac{(\xi - \frac{1}{6})^2}{8\omega_p^3}R^2 \\
& + \frac{(\xi - \frac{1}{6})m^2}{8\omega_p^5}(5\dot{R}H + 3R\dot{H} + 19RH^2) - \frac{25(\xi - \frac{1}{6})m^4}{16\omega_p^7}RH^2 \\
& + \frac{m^2}{16\omega_p^5}(\dot{H} + 15H\ddot{H} + 10\dot{H}^2 + 86H^2\dot{H} + 60H^4) \\
& - \frac{m^4}{32\omega_p^7}(28H\ddot{H} + 19\dot{H}^2 + 394H^2\dot{H} + 507H^4) + \frac{221m^6}{32\omega_p^9}H^2(\dot{H} + 3H^2) - \frac{1105m^8}{128\omega_p^{11}}H^4.
\end{aligned} \tag{32}$$